

Problem A: Consider the following measure spaces

$$(\Omega_i = \mathbb{N}, \mathcal{F}_i = \mathcal{P}(\mathbb{N}), \mu_i = \text{counting measure}), i = 1, 2.$$

Define $f : (\Omega_1 \times \Omega_2) \rightarrow \mathbb{R}$ as

$$f(\omega_1, \omega_2) = \mathbf{I}_{\{(n,n):n \in \mathbb{N}\}}(\omega_1, \omega_2) - \mathbf{I}_{\{(n,n+1):n \in \mathbb{N}\}}(\omega_1, \omega_2).$$

Then,

- (i) Prove $\mathcal{F}_1 \times \mathcal{F}_2 = \mathcal{P}(\mathbb{N} \times \mathbb{N})$.
- (ii) Is f a $\mathcal{F}_1 \times \mathcal{F}_2$ measurable function?
- (iii) Is f integrable w.r.t $\mu_1 \times \mu_2$?
- (iv) Compute the two iterated integrals:

$$\int_{\Omega_1} \left[\int_{\Omega_2} f(\omega_1, \omega_2) d\mu_2(\omega_2) \right] d\mu_1(\omega_1), \quad \int_{\Omega_2} \left[\int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1) \right] d\mu_2(\omega_2)$$